

A PROJECT TITLE
ON
Sports Injury Risk Prediction Using Random Forest and SVM
project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY
in
A.Y.: 2026-27

by

Name-G MANIDEEP PATEL

roll-2520090100

Name-R SAI CHARAN REDDI

roll-2520090184

Under the Esteemed guidance of

<Dr.Y. Lakshmi Prasanna>

<ASSISTANT PROFESSOR, CSIT>

<COMPUTER SCIENCE & INFORMATION TECHNOLOGY>



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

Name of Title: Sports Injury Risk Prediction Using Random Forest and SVM

1. Abstract:

should be 300 words

Sports injuries are a major concern for athletes, coaches, and sports organisations, as they can reduce performance, interrupt training schedules, and lead to long-term health problems. Early prediction of injury risk enables timely preventive measures, helping athletes maintain peak performance while reducing medical costs and recovery time. This project presents a machine learning-based approach for predicting sports injury risk using two widely adopted classification algorithms: Random Forest and Support Vector Machine (SVM). The objective is to develop an accurate and reliable prediction model that identifies athletes who are more likely to sustain injuries based on their physical, physiological, and training-related characteristics.

The proposed system uses a dataset containing features such as age, gender, body mass index (BMI), type of sport, weekly training hours, training intensity, previous injury history, fitness level, sleep duration, recovery time, and workload. The collected data undergoes preprocessing steps, including data cleaning, handling missing values, encoding categorical variables, feature scaling, and feature selection to improve model performance. The processed dataset is then divided into training and testing sets for model development and evaluation.

Random Forest is employed because of its ability to handle complex, non-linear relationships and reduce overfitting through ensemble learning, while Support Vector Machine is used for its effectiveness in binary classification and capability to find optimal decision boundaries. The performance of both algorithms is evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrix. A comparative analysis is performed to determine which algorithm provides better prediction accuracy and generalisation.

The expected outcome of this project is a predictive model capable of identifying athletes at high risk of injury before it occurs. Such a system can support coaches, trainers, physiotherapists, and sports physicians in designing personalised training programmes, monitoring athlete health, and implementing preventive strategies. Ultimately, the project demonstrates how machine learning techniques can contribute to enhancing athlete safety, improving performance, and reducing the occurrence of preventable sports injuries through data-driven decision-making.

2. INTRODUCTION:

Sports have become an essential part of modern life, promoting physical fitness, teamwork, and overall well-being. However, athletes are often exposed to the risk of injuries due to intense training, repetitive movements, overuse, inadequate recovery, and physical stress. Sports injuries not only affect an athlete's performance but can also lead to long recovery periods, increased medical expenses, and, in severe cases, the end of a sporting career. Therefore, identifying athletes who are at a higher risk of injury before an injury occurs has become an important area of research in sports science and healthcare.

Traditional methods of injury prevention mainly rely on physical examinations, medical history, and the experience of coaches or medical professionals. While these approaches are valuable, they may not always detect hidden patterns or complex relationships among multiple risk factors. With the rapid advancement of Artificial Intelligence (AI) and Machine Learning (ML), it is now possible to analyse large amounts of

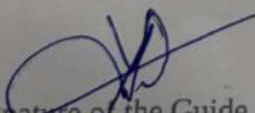
athlete data and predict injury risks more accurately. Machine learning algorithms can identify patterns within historical data, enabling early detection of potential injuries and supporting data-driven decision-making.

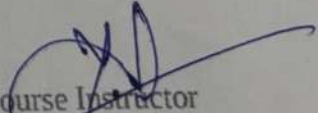
This project focuses on predicting sports injury risk using two popular supervised machine learning algorithms: Random Forest and Support Vector Machine (SVM). These algorithms are chosen because of their high accuracy, robustness, and effectiveness in solving classification problems. The prediction model is developed using athlete-related information such as age, gender, body mass index (BMI), training hours, training intensity, previous injury history, fitness level, sleep duration, recovery time, and workload. By analysing these factors, the model classifies athletes into different injury risk categories.

The proposed system aims to assist coaches, fitness trainers, physiotherapists, and sports physicians in identifying athletes who may be at a high risk of injury. Early prediction enables preventive measures such as personalised training plans, adequate recovery schedules, and workload management, thereby reducing injury occurrence and improving athletic performance. Overall, this project demonstrates how machine learning can enhance sports injury prevention by transforming historical athlete data into meaningful insights for safer and more effective training practices.

3. SOFTWARE REQUIREMENTS:

- ❑ **Operating System:** Windows 10/11, Ubuntu 20.04 or later, macOS
- ❑ **Programming Language:** Python 3.9 or above
- ❑ **Development Environment:** Jupyter Notebook / Google Colab
- ❑ **IDE:** Visual Studio Code or PyCharm
- ❑ **Machine Learning Library:** Scikit-learn
- ❑ **Data Processing Libraries:** Pandas, NumPy
- ❑ **Data Visualization Libraries:** Matplotlib, Seaborn
- ❑ **Dataset Format:** CSV (Comma-Separated Values)
- ❑ **Model Saving Library (Optional):** Joblib


Signature of the Guide


Course Instructor